

Pseudocode Exercises 1-26

1. Sum of List Elements

```
FUNCTION SumList(numbers)
    sum = 0
    FOR EACH number IN numbers
        sum = sum + number
    RETURN sum
END FUNCTION
```

2. Find the Largest Number

```
FUNCTION FindLargest(numbers)
    largest = numbers[0]
    FOR EACH number IN numbers
        IF number > largest THEN
            largest = number
    RETURN largest
END FUNCTION
```

3. Count Vowels

```
FUNCTION CountVowels(text)
    text = LOWERCASE(text)
    count = 0
    FOR EACH character IN text
        IF character='a' OR character='e' OR character='i' OR character='o' OR character='u' THEN
            count = count + 1
    RETURN count
END FUNCTION
```

4. Reverse a String

```
FUNCTION ReverseString(text)
    reversed = ""
    FOR i = LENGTH(text)-1 TO 0 STEP -1
        reversed = reversed + text[i]
    RETURN reversed
END FUNCTION
```

5. Check Palindrome

```
FUNCTION CheckPalindrome(text)
    text = REMOVE_SPACES(LOWERCASE(text))
    IF text = REVERSE(text) THEN
        RETURN True
    ELSE
        RETURN False
END FUNCTION
```

6. Factorial

```
FUNCTION Factorial(n)
    factorial = 1
    FOR i = 1 TO n
        factorial = factorial × i
    RETURN factorial
END FUNCTION
```

7. DazzBuzz

```
FUNCTION DazzBuzz()
    FOR number = 1 TO 100
        IF number MOD 4 = 0 AND number MOD 5 = 0 THEN PRINT 'DazzBuzz'
        ELSEIF number MOD 4 = 0 THEN PRINT 'Dazz'
        ELSEIF number MOD 5 = 0 THEN PRINT 'Buzz'
        ELSE PRINT number
    END FUNCTION
```

8. Find Prime Numbers

```
FUNCTION FindPrimes(start,finish)
    primes = EMPTY LIST
    FOR number = start TO finish
        IF number>1 THEN
            prime =True
            FOR i = 2 TO number-1
                IF number MOD i = 0 THEN prime = False ENDIF
            ENDFOR
            IF prime=True THEN ADD number TO primes ENDIF
        RETURN primes
    END FUNCTION
```

9. Fibonacci

```
FUNCTION Fibonacci(n)
    first=0
    second=1
    PRINT first
    PRINT second
    FOR i=3 TO n
        next=first+second
        PRINT next
        first=second
        second=next
    END FUNCTION
```

10. Calculate Area

```
FUNCTION CalculateArea(shape)
    IF shape='circle' THEN RETURN 3.14×radius×radius
    ELSEIF shape='rectangle' THEN RETURN length×width
```

```
ELSEIF shape='triangle' THEN RETURN (base×height)÷2
ELSE RETURN 'Invalid Shape'
END FUNCTION
```

11. Data Aggregation

```
FUNCTION TotalPrice(products)
totals=EMPTY DICTIONARY
FOR EACH product IN products
    language=product.language
    IF language IN totals THEN totals[language]=totals[language]+product.price ELSE
totals[language]=product.price RETURN totals
END FUNCTION
```

12. Even/Odd

```
FUNCTION CheckEvenOrOdd(number)
IF number MOD 2=0 THEN RETURN 'Even' ELSE RETURN 'Odd' ENDIF
END FUNCTION
```

13. Grade

```
FUNCTION CalculateGrade(mark)
IF mark>=90 THEN RETURN 'A'
ELSEIF mark>=80 THEN RETURN 'B'
ELSEIF mark>=70 THEN RETURN 'C'
ELSEIF mark>=60 THEN RETURN 'D'
ELSE RETURN 'F'
END FUNCTION
```

14. Sum Even

```
FUNCTION SumEven(numbers)
sum=0
FOR EACH number IN numbers
    IF number MOD 2=0 THEN sum=sum+number
RETURN sum
END FUNCTION
```

15. Find Smallest

```
FUNCTION FindSmallest(numbers)
smallest=number[0]
FOR EACH number IN numbers
    IF number<smallest THEN smallest=number ENDIF
RETURN smallest
END FUNCTION
```

16. Count Character

```
FUNCTION CountCharacter(text,ch)
count=0
FOR EACH c IN text
```

```
IF c=ch THEN count=count+1
RETURN count
END FUNCTION
```

17. Is Sorted

```
FUNCTION IsSorted(numbers)
FOR i←0 TO LENGTH(numbers)-2
IF numbers[i]>numbers[i+1] THEN RETURN False ENDIF
ENDFOR
RETURN True
END FUNCTION
```

18. Sort Asc

```
FUNCTION SortAscending(numbers)
SORT numbers ASCENDING
RETURN numbers
END FUNCTION
```

19. Sort Desc

```
FUNCTION SortDescending(numbers)
SORT numbers DESCENDING
RETURN numbers
END FUNCTION
```

20. Merge Lists

```
FUNCTION MergeLists(list1,list2)
RETURN list1+list2
END FUNCTION
```

21. Interest

```
FUNCTION CalculateInterest(p,r,t)
RETURN  $(p \times r \times t) \div 100$ 
END FUNCTION
```

22. Common Elements

```
FUNCTION CommonElements(list1,list2)
common←EMPTY LIST
FOR EACH item IN list1
IF item IN list2 THEN ADD item TO common ENDIF
ENDFOR
RETURN common
END FUNCTION
```

23. Celsius

```
FUNCTION ConvertTemperature(c)
RETURN  $(c \times 9 \div 5) + 32$ 
END FUNCTION
```

24. Sum Ten

```
FUNCTION SumTenNumbers()  
  sum←15  
  FOR i←1 TO 10  
    INPUT number  
    sum←sum+number  
  ENDFOR  
  RETURN sum  
END FUNCTION
```

25. Average

```
FUNCTION CalculateAverage()  
  sum←0  
  FOR i←1 TO 25  
    INPUT mark  
    sum←sum+mark  
  ENDFOR  
  RETURN sum÷25  
END FUNCTION
```

26. Login

```
FUNCTION Login(username,password)  
  IF username='admin' AND password='password123' THEN  
    RETURN 'Login Successful!'  
  ELSE  
    RETURN 'Invalid Credentials.'  
  ENDIF  
END FUNCTION
```